NANCY KAIM

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EDUCATION

Bachelor of Science in Computer Science Honours with Research CGPA: **8.091/10**
(Minor in Mathematics) *First Division*
University of Delhi, India Nov 2022 – June 2026

- **Coursework:** Artificial Intelligence, Machine Learning, Deep Learning, Information Security, Ethical Hacking, Advanced DSA, Software Engineering, Probability and Linear Programming.

RESEARCH EXPERIENCE

Optimisation of Internet of Things Technology in Military Domain Oct 2025 – May 2026
Undergraduate Dissertation, Supervisor: Prof. Archana Singhal *University of Delhi*

- Addressed performance decay and black-box opacity in ML-based Intrusion Detection Systems for IoT networks in high-stakes military settings, where unreliable models carry direct operational risk.
- Proposed a label-free selective retraining framework that triggers updates from model confidence drop and entropy rise, and compared it with static and blind periodic retraining across 6 classifiers (LR, DT, RF, KNN, XGBoost, SVM) on ToN_IoT.
- Achieved 89.86% mean accuracy with 13 updates, against 81.09% with 48 updates for blind retraining: **+8.77 points accuracy and 72.92% fewer retraining events**.
- Applied SHAP to track feature-importance stability across drift windows, showing that some models stay misleadingly stable while accuracy drops, so explainability should be monitored dynamically.
- **Best Paper Award (IoT Track)** at *International Conference on Emerging Trends in AI & CS* 2026

PUBLICATIONS

[1] Verma, T., Rajak, M., **Kaim, N**., Sharma, N., Nisha, & Singh, H. P. (2025, December). Bridging Gaps in IoT Literature: A Thematic Categorization of Challenges and Resolutions. In *International Conference on Information and Communication Technology for Competitive Strategies* (pp. 197-209). Cham: **Springer Nature** Switzerland.

[2] Rajak, M., **Kaim, N.**, Sharma, N., Verma, T., Singh, H. P., & Nisha. (2026). The Evolving Paradigm of IoT: A Systematic Literature Survey of Challenges, Solutions, and Emerging Technologies. *Blockchain Solutions for Securing Internet of Things Networks*, 1-37. (**Wiley**)

TECHNICAL PROJECTS

- **AI-Based Interactive Chatbot | Smart India Hackathon Prototype:** Developed a modular, domain-specific Chatbot prototype using fine-tuned GPT-2 and query routing to address queries on judicial services, eCourts, and legal aid, with extensibility for future legal-information services.
- **AI Chat Interface for Instant Insights:** Built and deployed a Streamlit-based natural-language interface integrating Google's TAPAS model to query structured datasets conversationally, enabling data retrieval without requiring users to write SQL queries.
- **Smart Plant Pot | IoT-Based Water Management:** Engineered an Arduino-based soil-moisture monitoring prototype integrating a Node.js server and WebSocket communication to deliver real-time browser updates, enabling responsive plant-watering decisions and supporting water conservation.

SKILLS

Programming: Python, SQL, JavaScript
Machine Learning & Explainability: LLM, Scikit-learn, PyTorch, RAG, Transformer, SHAP

Research: Intrusion Detection Systems (IDS), Concept Drift, Explainable AI
Development Tools: Streamlit, Node.js, Arduino, Git/GitHub